Analyse IV - Summary

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2026

1 Information

A concise and rigorous summary transcribing the most important concepts and theorems from the Analyse IV lecture at EPFL.

2 Holomorphic and Analytic Functions

2.1 Definitions

Let $\Omega \subseteq \mathbb{C}$ be an open set. A complex function $f : \Omega \to \mathbb{C}$ is said to be **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

If f is complex differentiable at *every* point of Ω, we say f is **holomorphic** on Ω.
If f is holomorphic on the entire complex plane ($\Omega = \mathbb{C}$), we call f an **entire function** (e.g., polynomials, e^z, $\sin(z)$).

3 The Cauchy-Riemann Equations

Let $z = x + iy \in \mathbb{C}$. A complex function $f(z)$ can be separated into its real and imaginary parts:

$$f(x + iy) = u(x, y) + iv(x, y)$$

where $u, v : \mathbb{R}^2 \to \mathbb{R}$.

3.1 Necessary and Sufficient Conditions

Theorem: Suppose $u(x, y)$ and $v(x, y)$ have continuous first-order partial derivatives (i.e., they are of class C^1) on an open set Ω. The function $f = u + iv$ is holomorphic on Ω if and only if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

When these equations hold, the complex derivative can be computed as:

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

3.2 Harmonic Conjugates

Because holomorphic functions are infinitely differentiable, u and v have continuous second-order derivatives. By applying the Cauchy-Riemann equations and Schwarz's theorem (symmetry of second derivatives), we get:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Therefore, the real and imaginary parts of a holomorphic function are **harmonic functions**. We say that v is the harmonic conjugate of u.

4 The Complex Logarithm

4.1 Principal Argument

For any $z \neq 0$, the principal argument $\text{Arg}(z)$ is the unique angle in the interval $(-\pi, \pi]$. It is computed using the arctangent function depending on the quadrant of (x, y).
Crucial limitation: $\text{Arg}(z)$ is discontinuous across the negative real axis.
this is how we compute it :

$$\text{Arg}(z) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \end{cases}$$

4.2 Principal Branch of the Logarithm

We define the **principal branch** of the complex logarithm, denoted $\text{Log}(z)$, as:

$$\text{Log}(z) = \ln|z| + i\text{Arg}(z) = \ln\left(\sqrt{x^2 + y^2}\right) + i\text{Arg}(x + iy)$$

Rigorous Properties:

- **Domain:** $\text{Log}(z)$ is a well-defined, single-valued function on the cut plane $\mathbb{C} \setminus (-\infty, 0]$. We must remove zero and the negative real axis (the branch cut) to maintain continuity.
- **Holomorphy:** On this cut plane, $\text{Log}(z)$ is holomorphic.
- **Derivative:** Its complex derivative is exactly what we expect from real analysis:

$$\frac{d}{dz}\text{Log}(z) = \frac{1}{z} \quad \text{for } z \in \mathbb{C} \setminus (-\infty, 0]$$